The trials follow only households already in a bind: adults who are homeless, most with SMI and many with a SUD. They do not follow the households that could get into a bind but that the condition deters from starting to use drugs. Nor do they observe the vouchers the condition's savings buy for others, or the children's adult lives. Once those are counted, aggregate homelessness and childhood homelessness drop and welfare increases.

Third, homelessness and childhood homelessness do not fall along the transition if treatment and sobriety are not a real option for the households that already have a SUD. A temporary exemption for those with a SUD keeps the latter from ever rising and the former's rise small: if a household in treatment keeps its voucher, sober or not, for the first five periods, and the screen catches one in five users, the share of children in a shelter never rises above its Housing First level and homelessness rises by at most 1.5 percent, with newborns gaining even more than in the main transition.

A Calibration of the standard parameters

The remaining thirty-three parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Six of them I calibrate internally, against a moment, because no source gives them directly; the other twenty-seven I set from outside, from an estimate or from statute. Table A1 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the six I calibrate internally, since the other twenty-seven have no target to reproduce.

The parameters in this appendix are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of productivity determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters of Section 5 is not that they do nothing, but that they are not specific to this paper.

Table A1: Standard parameters

Parameter Description Set Moment and source Data Model Value

Block 7: Preferences σ Risk aversion Ext. Attanasio and Weber (2010) 2.00 n.a. 2.00 γ Weight on housing services Int. Housing over expenditure net of drugs, CEX 2024 26.8% 26.2% 0.536 β Discount factor Int. Wealth over output, BEA fixed assets 3.00 2.95 0.946 ξ(j) Equivalence scale Ext. Square-root scale, OECD 1.41 n.a. 1.41

Block 8: Rental ladder h2 Smallest rental size Int. Renters paying over half of income, SCF 2022 19.2% 25.8% 0.0407 h3/h2 Middle rung over the smallest Ext. Set 4.00 n.a. 4.00 h4/h2 Largest rung over the smallest Ext. Set 8.00 n.a. 8.00

Block 9: Earnings and employment J Maximum age Ext. Age 85 60 n.a. 60 JK Years the child lives at home Ext. Parent aged 26 to 50 25 n.a. 25 JR Retirement age Ext. Age 66, SSA full retirement age 41 n.a. 41 ε(j) Age profile of efficiency Ext. Hump peaking at age 56, Greenwood et al. (2025) Profile n.a. Profile ρy Persistence of productivity Ext. Guvenen

et al. (2023) 0.90 n.a. 0.90 σζ Standard deviation of its shocks Int. Bottom half's share of earnings, SCF 2022 17.2% 16.2% 0.413 ρ Persistence of productivity across generations Ext. Solon (1992), Mazumder (2005) 0.50 n.a. 0.50 ¯πeu Job separation rate Int. Unemployment rate, BLS CPS 2014–24 4.9% 5.6% 0.0268 ϖ(y) Separations by productivity Ext. Unemployment by education, BLS 2019 Profile n.a. Profile ¯πue Annual job-finding rate Ext. Krueger et al. (2014) 0.55 n.a. 0.55 pui Share of separations that receive UI Ext. UI applicants among the unemployed, BLS 2022 0.30 n.a. 0.30

Block 10: Baseline survival and SMI onset

¯ϕ(j) Baseline survival by age Ext. US life table 2010, Arias (2014) Profile n.a. Profile ςm(j) Age profile of SMI onset Ext. SMI prevalence by age, NSDUH 2024 Profile n.a. Profile

Block 11: Interest and taxes r Risk-free rate Ext. Kaplan et al. (2020) 0.03 n.a. 0.03 φ Tax progressivity Ext. Heathcote et al. (2017) 0.85 n.a. 0.85 τℓ Level of the labour income tax Int. Average labour income tax, McDaniel (2007) 22.4% 20.9% 0.894 τk Capital income tax rate Ext. US long-run rate on gains and dividends 0.15 n.a. 0.15

Block 12: Social security benefits and programmes' rules S(y) Social security Ext. SSA primary insurance amount Statute n.a. 0.90, 0.32, 0.15 S SSI floor over average labour income Ext. SSA federal benefit rate 0.177 n.a. 0.177 ψui UI replacement rate Ext. Department of Labor, statutory rates 0.50 n.a. 0.50 ¯y Cap on the UI earnings base Ext. Department of Labor, state benefit maxima 1.00 n.a. 1.00 κV Voucher means test Ext. HUD rule 0.50 n.a. 0.50 κr Tenant's rent, share of income Ext. HUD rule 0.30 n.a. 0.30 ϱ Food stamp clawback rate Ext. SNAP statute, 7 CFR 273.10 0.30 n.a. 0.30

Block 13: Drug market JE Maximum age at which use can begin Ext. Model age 55, that is age 80 55 n.a. 55 ν1 Supply shifter Ext. Normalisation, so pv = 1 Norm. n.a. Norm.

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 7: Preferences. The four parameters in this block determine what a household wants, and what it is willing to give up to get it.

σ, risk aversion. It sets the marginal utility of consumption, so it affects every welfare comparison in

the paper. Increase it and households want smoother consumption and insurance is worth more, the voucher included. I set it to the standard value of two, following Attanasio and Weber (2010).

γ, the weight on housing services. It is the weight on housing in the bundle of consumption and housing,

so it sets what a dwelling is worth against consumption, and that bundle is what the drug is weighed against. As γ rises housing is worth more relative to consumption, households buy more of it, and the share of spending that goes on housing rises with it. γ is chosen for the share it produces: shelter plus utilities take 26.8 percent of expenditure in the 2024 Consumer Expenditure Survey (CEX), which excludes illicit drug spending. β, the discount

factor. It sets how much households discount the future, and thus how much they save

and how much the prospect of death deters them from using drugs. As it rises households save more, the wealth-to-output ratio rises and households are more deterred by death. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover. ξ(j), the equivalence scale. The household's consumption and dwelling are shared with the child while

it lives with its parent, and thus are divided by ξ(j) to give what each member gets. Raise it and each member gets less of a given budget, and the child counts for more of the household, through the weight $\xi(j) - 1$ in (12). It is the square-root scale (OECD, 2013),

$\sqrt{}$ N(j), where N(j) in the model is set to two while the child lives with the parent ($j \leq JK$), from the parent's age 26 to 50, and to one after the child has left ($j > JK$).

Block 8: Rental ladder. These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, h1, which was calibrated in Block 2.

h2, the smallest rental size. The size of the cheapest dwelling is what decides how much of its income a

poor household must spend to be housed at all. Raise it and the bottom rung is larger and dearer, so more households are burdened, and a voucher is worth more to them. It is chosen so that the fraction of renters paying more than half of their income for housing is similar to the data, 19.2 percent in the 2022 SCF; the model gives 25.8 percent. h3/h2 and h4/h2, the spacing of the ladder. These two ratios set how much housing the middle and

largest rungs provide, and so what a household that can afford more than the smallest unit can move up to. Raise them and each step up the ladder costs more rent, and fewer households can afford the larger rungs. The two larger rungs are set at 4.00 and 8.00 times the smallest.

Block 9: Earnings and employment. These eleven parameters determine how long a household lives and works, what it is expected to earn over its life, how persistent the shocks to it are, and how it moves between work and unemployment.

J, JK and JR, the length of life, of child rearing and of working life, respectively. A household lives

at most J periods, has its child at home for its first JK of them and works until model age JR, after which it retires on social security. Together they set how many years of life and of earnings a disorder can cost, and JK how long a parent's drug use reaches its child through what they share. Raise either J or JR and a disorder that shortens life or work costs more; raise JK and the budget and childhood homelessness externalities last longer. I set $J = 60$, so that with $j = 1$ at age 26 the last age is 85, $JK = 25$, so that the child lives at home while its parent is 26 to 50, and $JR = 41$, so that retirement begins at age 66, the full retirement age under social security for those born between 1943 and 1954. ε(j), the age profile of efficiency. This is the deterministic hump in earnings over the life cycle.

Without

it every household with the same productivity would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life. I take it as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age. ρ_y and σ_ζ, the productivity process. Log productivity is persistent but not permanent: it follows an

autoregressive process with persistence ρ_y and shocks with standard deviation σ_ζ. σ_ζ is what decides how many households are poor enough to be at risk of homelessness, and the share of earnings that

goes to the bottom half of working households falls as it rises. I set $\rho_y = 0.90$, the persistence of the income process in Guvenen et al. (2023). σ_ζ is calibrated to the bottom half's share of earnings among households aged 26 to 65 with positive wage income in the 2022 SCF, 17.2 percent. ρ, the persistence of productivity across generations. The child's log productivity at entry into adult-